Circuit model and solder practice

antocciphysics | TEACHER KEY | CPA

Learning target: Test a circuit representation before making permanent connections.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 17; 3 course days later.*
Check circuit/secured parts/clear wheels with teacher approval. Measure distance and elapsed time or another planned outcome.

02. Explain a worked example (worked)
One loop through the motor; switch interrupts it; no direct low-resistance bypass across battery.

03. Look and explain (model)
Closed loop permits current; open switch interrupts the conducting path.

04. Build the solution (practice)
.60 W; simulation observation must come from the student’s trial.

05. Use the idea (reason)
Student records actual feedback, such as secure connection, wetting or excess solder; no invented approval.

06. Choose a method (mixed) *Review: Unit 7, Day 8; 12 course days later.*
1 A, by charge conservation.

07. Exit check (exit)
Disconnected; avoid unintended current, shorts and powered motion during work.

08. STRETCH: Apply it another way (stretch)
Open path gives no steady current; bypass risks excessive current. Disconnect power, inspect continuity/routing under classroom procedure.

Teaching note: Record actual practice and teacher feedback. Do not energize a circuit while soldering or handling exposed connections.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 17; 3 course days later.*
20 J may go to wheel rotation, heating, friction, drag or sound. Total energy accounting includes these terms.

02. Explain a worked example (worked)
One loop through the motor; switch interrupts it; no direct low-resistance bypass across battery.

03. Represent and interpret (model)
Closed loop permits current; open switch interrupts the conducting path.

04. Solve and justify (practice)
20 Ω; 1.8 W; motor rotation generates back emf and behavior depends on load and speed.

05. Analyze the reasoning (reason)
Student records actual feedback, such as secure connection, wetting or excess solder; no invented approval.

06. Choose a method (mixed) *Review: Unit 7, Day 8; 12 course days later.*
Parallel 6 Ω; series 24 Ω.

07. Exit check (exit)
Appearance alone does not establish wetting, continuity or mechanical strength; use approved inspection/testing.

08. STRETCH: Apply it another way (stretch)
Running 2 A, 12 W; stalled 6 A, 36 W. Effective motor behavior includes back emf; do not intentionally sustain a stall.

Teaching note: Record actual practice and teacher feedback. Do not energize a circuit while soldering or handling exposed connections.